

S^n , ~~S^1~~ , RP^n , $\langle \dots \rangle$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$, 2-dim mfd, $L(p,q)$

$M \cup N$, $\wedge \vee$, $\langle X, S \rangle / \langle e^{2\pi i}, e^{2\pi i} \rangle$
 π_0, π_1^2 , Kleim., free product, $\langle a, b \mid aba = b \rangle$, $ababab...$
 $\mathbb{Z} \times \mathbb{Z}$, $abab^t = e$, $\langle a, b \mid aba = b \rangle$, $ab = ba \rightarrow$ group $\mathbb{Z} \times \mathbb{Z}$
 $\frac{S \vee S'}{\mathbb{Z} \times \mathbb{Z}}$, $T: \mathbb{Z} \times \mathbb{Z}$, $T \in \mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$ theorem

Theorem: $M: n$ -dim with $(n \geq 2)$
 then $\pi_1(M) \cong \pi_1(M \setminus \{x\})$
 $\pi_1(RP^n) \cong \mathbb{Z}_2$ ($n \geq 2$), $\pi_1(M \setminus \{x\}) \cong \pi_1(M \setminus \{x, y\})$
 $\pi_1(S^n) = 0$ ($n \geq 2$), $R^3: 0$

$\langle \dots \rangle$, $\langle S, S' \vee R \rangle$, $\mathbb{Z} \times \mathbb{Z}$
 $\frac{S \vee S'}{\mathbb{Z} \times \mathbb{Z}}$
 $S^2/p = S$, S^2 : 0-cell $\mathbb{Z}$, 1-cell $\mathbb{Z}$, 2-cell $\mathbb{Z}$
 R^2 : 0-cell: $\mathbb{Z}$, 1-cell: $\mathbb{Z}$, 2-cell: $\mathbb{Z}$

Example: $\pi_1(M) \cong \pi_1(M \setminus \{x\})$, $\pi_1(M \setminus \{x, y\})$
 $\pi_1(RP^n) \cong \pi_1(RP^n \setminus \{x\}) \cong \mathbb{Z}_2$
 $\pi_1(RP^n) \cong \pi_1(RP^n \setminus \{x, y\}) \cong \mathbb{Z}_2$

R^2 : 0-cell: $\mathbb{Z}$, 1-cell: $\mathbb{Z}$, 2-cell: $\mathbb{Z}$
 $R^3: \mathbb{Z}_2$, $\langle a \mid a^2 \rangle$, $a^2 = 1$
 $\pi_1(RP^n) \cong \pi_1(RP^n \setminus \{x\}) \cong \mathbb{Z}_2$
 $\pi_1(RP^n) \cong \pi_1(RP^n \setminus \{x, y\}) \cong \mathbb{Z}_2$

$\mathbb{R}/\mathbb{Z}$, $\mathbb{Z} = \mathbb{Z} + i\mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$

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